Response to the Editor and Referee

Manuscript: Crowded Out: Measuring Rental Market Pressure with Household Density

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Dear Professor Ambrose,

Thank you for continuing the editorial process and for the referee's careful and constructive engagement with our revision. We are grateful that Referee 2 finds merit in the paper. Below we provide a point-by-point response to each of the three remaining comments. We believe the revision addresses all three concerns fully, and we hope the referee will agree.

Comment 1: *"I had suggested in my previous report that renter population per bedroom would be a more appropriate measure of excess demand than renters per unit. The author has done something different with this insight. An 'equilibrium' RDI is created, which is a function of the age distribution of the population and the number of bedrooms per renter. So in effect the author is regressing renters per unit on renters per bedroom, which looks a lot like a regression of (the inverse of) bedrooms per unit, so it's no wonder that the coefficient looks like the number of bedrooms per unit (2.4). The author doesn't really explain this model and I am not sure what it is telling us, but it may not be telling us very much."*

Response: The referee's concern is well-taken, and we agree that the prior approach was difficult to interpret and potentially circular. In this revision, we have abandoned the two-stage "equilibrium RDI" construction entirely. We no longer regress renters per unit on renters per bedroom to generate a residual. Instead, we incorporate bedroom density (renters per rented bedroom) directly as a control variable in a single-stage panel ARDL specification (Section 4.1, Table 4). This resolves the circularity the referee identified: bedroom density now enters as an explicit covariate that absorbs persistent compositional differences across cities. This term accounts for metropolitan areas where larger households share more bedrooms by family preference rather than market pressure. And, it is no longer observed in the left-hand-side variable. With bedroom density held fixed, the RDI level term captures something distinct and interpretable: within-metro variation over time in how many people share each unit, net of structural household composition. When a metropolitan area becomes more crowded than its own history and fixed characteristics would predict, controlling for bedroom-level composition and supply changes, that excess density signals accumulated demand pressure that has not yet been resolved through prices. The coefficient on the RDI level ($\lambda = 0.0463$, $p < 0.01$) reflects this: a one-standard-deviation increase in RDI is associated with 130 basis points of additional real rent growth in the following year. This is not a mechanical relationship between bedrooms and units; it is a forward-looking signal about market tightness operating through household formation behavior. The coefficient on bedroom density in the ARDL ($\kappa = -0.0846$, $p < 0.05$) has a complementary and intuitive interpretation: markets where renters are structurally more crowded per bedroom, due to family formation patterns rather than scarcity, exhibit lower subsequent rent growth once aggregate demand intensity is held fixed, consistent with households already at their space-consumption limit. The opposing signs on RDI and bedroom density are therefore not contradictory but complementary: RDI captures dynamic demand pressure within markets over time, while bedroom density absorbs persistent compositional differences across markets. The RDI level and change terms now stand on their own as predictors of rent growth, without the generated-regressor concerns that attended the prior approach.

Comment 2: *"In order to avoid confusion the author should note that this is not really an error correction model as that term is usually understood. That model is generally of the form dy(t) = dy(t-1) + dx(t-1) + (y-x)(t-1) (coefficients omitted). This model is dy(t) = dy(t-1) + (x-z)(t-1). The dependent variable is normally part of the error calculation. It's not like the present formulation is wrong, but it is non-standard."*

Response: The referee is correct, and we thank them for the clarifying note. We have revised the terminology throughout the paper. The specification is now consistently described as a panel Autoregressive Distributed Lag (ARDL) model rather than an error correction model. Where we describe the role of the lagged level of RDI, we now explicitly characterize it as playing a role *analogous* to an error-correction term. It serves to capture accumulated demand pressure. We also note that the specification is a stationary ARDL rather than a cointegrating ECM in the Engle-Granger sense. Specifically, Section 4.1 now reads: "The specification follows the ARDL representation of an error-correction model. The lagged level $RDI_{\_m,t}$ plays a role analogous to an error-correction term in a traditional ECM, capturing accumulated demand pressure, though we note that because the series are stationary rather than integrated, this is properly understood as a stationary ARDL rather than a cointegrating relationship." The same clarification is carried into the stock-flow motivation in Section 4.3 and wherever ECM language previously appeared. The abstract and conclusion has also been updated accordingly.

Comment 3: *"Nevertheless, the equilibrium adjustment model is not estimated correctly. In the classic Engle-Granger error correction model estimation, it is correct to estimate the (long run) equilibrium relationship in a first stage, and place the error in the dynamic model in the second. This works only because the variables are integrated (of order 1) and the estimates of the first stage are super consistent. I don't think this data is integrated — you could of course do the panel ADF test, but it seems unlikely — so this doesn't work. The* *equilibrium error has all the problems of a generated regressor. The model should be estimated in a single step, basically regressing the rent change on the lagged changes and the lagged levels."*

Response: This comment identifies the most important econometric issue with the prior version, and we have addressed it fully and directly. Following the referee's recommendation, we now estimate the model in a single step. The revised specification (Equation 14, Section 4.1) regresses next-year real rent growth on the lagged level of RDI, the contemporaneous change in RDI, the lagged level and change of supply growth, lagged rent growth, and bedroom density, all within a panel ARDL framework with metropolitan and year fixed effects. This is precisely the single-stage approach the referee recommends.

We also directly test the stationarity assumption the referee raises. Using an Im-Pesaran-Shin-style panel unit root test we obtain t-statistics of −3.07 for $RDI_{m,t}$ and −3.25 for $\Delta R_{m,t+1}$, both below the 5% critical value of −2.98. This confirms that both series are stationary, consistent with the referee's intuition that the data are unlikely to be cointegrated. As the referee notes, this means the super-consistency justification for two-stage Engle-Granger estimation does not apply, and a single-stage approach is both appropriate and necessary. We state this explicitly in Section 4.1: "Single-stage estimation is therefore appropriate and avoids the generated-regressor problems that would arise from a two-stage approach. This also confirms that the lagged level of RDI enters as a stationary predictor rather than a cointegrating vector, so the specification is an ARDL rather than an error-correction model in the Engle-Granger sense."

The shift to single-stage estimation yields meaningful improvements in model fit. The within-R^2 rises to 0.141 from the lower values in the prior specification, and Ljung-Box tests at lag lengths 1 through 4 find no evidence of residual autocorrelation (lag 1: p = 0.458; lag 2: p = 0.676; lag 3: p = 0.785; lag 4: p = 0.111), confirming that the dynamic structure is well-specified.

We believe these revisions fully address the referee's remaining concerns. We feel the paper is stronger for them. The shift to single-stage ARDL estimation raises the within-R^2 to 0.141, and Ljung-Box tests at lags 1–4 find no evidence of residual autocorrelation, confirming the dynamic structure is well-specified. We are grateful for the referee's expert econometric guidance and hope this version is suitable for publication.